

# Across Fusion and Diversity Improves Mission-Oriented Image Transmissions in UAV-Enabled Wireless Networks

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**Abstract**—This letter proposes a mission-oriented semantic communication framework for image transmissions in unmanned aerial vehicle enabled wireless networks, targeting object localization. To improve the effective transmission rate, only task-relevant semantics, including target bounding-box coordinates, cropped object images, and background indices, are delivered, while non-critical background content is recovered from a shared knowledge base. A three-layer across-fusion-aided YOLO (You Only Look Once) model extracts spatially consistent object-centric semantics via multi-scale feature interaction. Maximal-ratio combining is further exploited in the critical data transmission to enhance reliability. Simulation results over Rician fading channels substantiate improved compression efficiency, reconstruction quality, and robustness over benchmarks.

**Index Terms**—Across Fusion (AF), knowledge base, maximal-ratio combining (MRC), semantic communication (SemCom), unmanned aerial vehicle (UAV), wireless networks, YOLO.

## I. INTRODUCTION

Unmanned aerial vehicles (UAVs) have been widely adopted in emergency rescue, environmental monitoring, agricultural inspection, and intelligent surveillance, owing to their flexible deployment, high mobility, and wide-area sensing capability [1]. In object localization tasks, UAVs are required to capture visual observations and transmit task-relevant information to a base station (BS) for inference and decision-making. However, conventional image transmission schemes convey high-resolution visual data at the pixel or bit level, incurring substantial communication overhead and imposing a heavy burden on bandwidth-limited UAV networks.

Semantic communication (SemCom) has emerged as a promising paradigm to improve communication efficiency by transmitting task-relevant semantic information rather than raw source data [2]. Unlike traditional systems that pursue symbol-level fidelity, SemCom preserves the semantic meaning required by downstream tasks, making it particularly suitable for mission-oriented UAV image transmissions where the objective is reliable recovery of task-critical content instead of pixel-wise reconstruction.

Existing image SemCom research can be broadly categorized into two paradigms: unimodal semantic representation

learning and multimodal feature fusion. The former focuses on designing deep neural networks to extract compact visual features that enable efficient transmission [3]. The latter leverages both textual and visual information to enrich semantic representations and enhance reconstruction fidelity [4]. Despite these advances, two key challenges remain, particularly in object localization tasks [5]: *i)* the spatial positioning of targets is difficult to preserve under semantic compression, often resulting in distorted reconstructions; and *ii)* both critical semantics and background regions are vulnerable to wireless channel corruption, which undermines visual quality and impairs task-level consistency.

To address these challenges, we propose a mission-oriented SemCom framework for UAV-enabled wireless networks. The proposed framework transmits only essential semantic information, including the target coordinates, cropped object image, and background image index, while recovering non-critical background regions from a shared knowledge base, significantly improving compression ratio and effective transmission rate. A three-layer across-fusion (AF) aided YOLO model is developed to enhance object localization accuracy via multi-scale semantic-spatial feature interactions, and maximal-ratio combining (MRC) is incorporated to ensure reliable coordinate transmission over wireless channels.

The main contributions of this letter are summarized below:

- A mission-oriented SemCom framework is proposed for UAV-enabled wireless networks, which transmits only task-relevant semantics and replaces redundant background data with a knowledge-base index, thereby significantly improving compression ratio and effective transmission rate.
- A three-layer AF-aided YOLO model is designed to extract object-centric semantic features, enhancing spatial localization accuracy and reconstruction fidelity.
- MRC is incorporated into the transmission of critical coordinate streams over Rician fading channels, providing diversity gain and improving both communication reliability and end-to-end semantic reconstruction robustness.
- A background image knowledge base is constructed using *Stable Diffusion Inpainting*, which supplies reliable background priors at the receiver, mitigating background distortion and boosting reconstruction performance in wireless channels.

The remainder of this letter is organized as follows. Sec-

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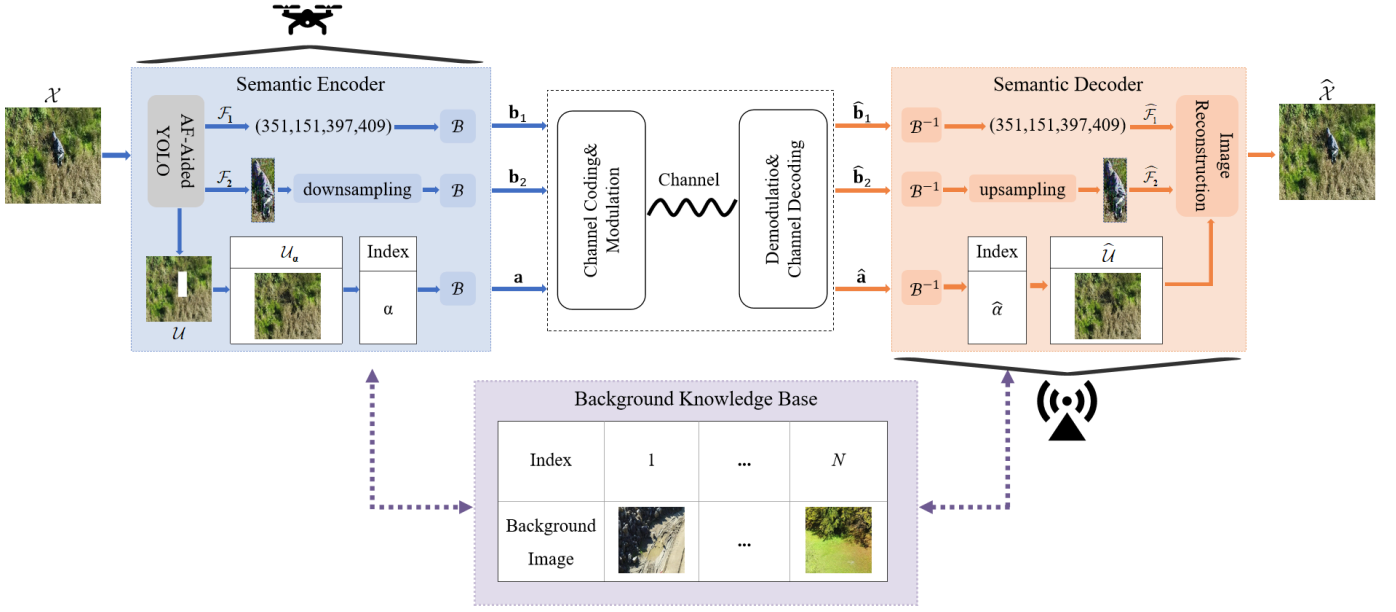


Fig. 1 The proposed SemCom system for image transmissions in a UAV-enabled wireless network.

tion II presents the proposed SemCom system with MRC in UAV-enabled wireless networks. Section III introduces the key modifications of our AF-aided YOLO model. Section IV evaluates the performance of the proposed framework and compares it with representative benchmarks. Section V concludes this letter.

## II. SEMCOM FRAMEWORK

Consider a SemCom system for image transmissions within a UAV-enabled wireless network, as illustrated in Fig. 1. The UAV observes an outdoor target object and transmits the extracted semantic information to the BS over a wireless channel. Moreover, the UAV is equipped with a semantic encoder for task-relevant feature extraction, whereas the BS employs a semantic decoder for image reconstruction. In addition, a shared background image knowledge base is maintained at both the UAV and the BS to facilitate semantic reconstruction and reduce redundant background transmission.

### A. Wireless Fading Model

Given the existence of both line-of-sight (LoS) and non-line-of-sight (NLoS) components in UAV-ground links, the wireless channel is modeled as a Rician block fading channel. The channel coefficient remains constant over a block of  $L$  consecutive symbols and changes independently across blocks. The channel coefficient for each block is expressed as

$$h = \sqrt{\frac{\kappa}{\kappa + 1}} h_{\text{LoS}} + \sqrt{\frac{1}{\kappa + 1}} h_{\text{NLoS}}, \quad (1)$$

where  $\kappa$  is the Rician factor and  $h_{\text{LoS}}$  denotes the deterministic LoS component. The scattered NLoS component  $h_{\text{NLoS}} \sim \mathcal{CN}(0, 1)$  is independently drawn for each fading block.

### B. Mission-Oriented Semantic Encoder

The source image is denoted by a third-order tensor  $\mathcal{X} \in \mathbb{R}^{H \times W \times C}$ , where the first two dimensions represent the image's height and width (pixel positions), whereas the third dimension represents the color channels. Instead of transmitting the entire image, the proposed semantic encoder extracts three task-relevant semantic components:

$$\mathbf{F} = \{\mathcal{F}_1, \mathcal{F}_2, \alpha\}, \quad (2)$$

where  $\mathcal{F}_1$  contains the bounding-box coordinates of the target object,  $\mathcal{F}_2$  denotes the cropped object image, and  $\alpha$  denotes the background image index.

Specifically,  $\mathcal{F}_1$  is extracted by our three-layer AF-aided YOLO detector, denoted by  $\mathcal{E}_1(\cdot)$  with model parameters  $\theta_1^*$  and introduced in Section III, expressed as

$$\mathcal{F}_1 = \mathcal{E}_1(\mathcal{X}|\theta_1^*) = (\xi_1, \eta_1, \xi_2, \eta_2), \quad (3)$$

where  $(\xi_1, \eta_1)$  and  $(\xi_2, \eta_2)$  represent the upper-left and lower-right coordinates of the detected object region. Guided by the detected bounding box,  $\mathcal{F}_2 = \mathcal{E}_2(\mathcal{X}|\theta_2^*)$  is obtained through the cropping and downsampling operation  $\mathcal{E}_2(\cdot)$  with model parameters  $\theta_2^*$ . To construct the background image index, the detected object region in  $\mathcal{X}$  is masked to obtain a object-removed background image  $\mathcal{U}$ . Given the shared background image knowledge base  $\mathcal{U} = \{\hat{\mathcal{U}}_1, \hat{\mathcal{U}}_2, \dots, \hat{\mathcal{U}}_N\}$ , the background index is obtained by closest-neighbor matching:

$$\alpha = \arg \min_{i \in \{1, \dots, N\}} \|\mathcal{U} - \hat{\mathcal{U}}_i\|_F^2, \quad (4)$$

where  $\hat{\mathcal{U}}_i$  is the  $i$ -th reference background image in the knowledge base. The background images in the set  $\mathcal{U}$  are generated offline by *Stable Diffusion Inpainting*, which provides diverse

and semantically consistent background priors for outdoor scenarios.

These extracted semantic components are then mapped into bit streams as

$$\mathbf{b}_1 = \mathcal{B}(\mathcal{F}_1), \quad \mathbf{b}_2 = \mathcal{B}(\Lambda(\mathcal{F}_2)), \quad \mathbf{a} = \mathcal{B}(\alpha), \quad (5)$$

where  $\mathcal{B}(\cdot)$  denotes binary mapping and  $\Lambda(\cdot)$  denotes the downsampling operation. The overall transmitted semantic bit stream is  $[\mathbf{b}_1, \mathbf{b}_2, \mathbf{a}]$ .

#### C. Transmission with Maximal-Ratio Combining

After channel coding and modulation, the transmitted symbol sequence is written as  $[\mathbf{s}_1, \mathbf{s}_2, \mathbf{c}]$ , where  $\mathbf{s}_1$ ,  $\mathbf{s}_2$ , and  $\mathbf{c}$  denote the modulated symbol vectors pertaining to the bit vectors  $\mathbf{b}_1$ ,  $\mathbf{b}_2$ , and  $\mathbf{a}$ , respectively.

Since the cropped object image usually dominates the semantic payload, the length of  $\mathbf{s}_2$  is much larger than that of the critical side-information stream  $[\mathbf{s}_1, \mathbf{c}]$ , which contains the bounding-box coordinates and the background image index. Given that the coherence length of the block-fading channel is  $L$  symbols, the transmission of  $\mathbf{s}_2$  generally spans multiple fading blocks. In contrast, the side-information stream  $[\mathbf{s}_1, \mathbf{c}]$  incurs a negligible signaling overhead but is crucial for preserving the spatial consistency of the reconstructed image. Therefore, to enhance the reliability of this critical semantic stream, it is repeatedly transmitted over  $M$  independently fading blocks and combined at the receiver using MRC.

The received signal in the  $m$ -th fading block is given by

$$\mathbf{y}_m = h_m [\mathbf{s}_1, \mathbf{c}] + \mathbf{n}_m, \quad m = 1, 2, \dots, M, \quad (6)$$

where  $h_m$  denotes the Rician channel coefficient of the  $m$ -th block, and  $\mathbf{n}_m \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I})$  is the additive white Gaussian noise vector. With perfect channel state information at the receiver, the MRC output is obtained as  $\tilde{\mathbf{y}} = \sum_{m=1}^M h_m^* \mathbf{y}_m$ , where  $(\cdot)^*$  denotes the complex conjugate. The combined signal  $\tilde{\mathbf{y}}$  is then passed to the demodulator and channel decoder to recover the critical semantic bits  $[\hat{\mathbf{b}}_1, \hat{\mathbf{a}}]$ . Meanwhile, the cropped object image bit vector  $\hat{\mathbf{b}}_2$  is obtained from the received fading blocks carrying  $\mathbf{s}_2$  through conventional demodulation and channel decoding.

#### D. Semantic Decoder

At the BS, the semantic decoder reconstructs the image from the received semantic components. The coordinate and cropped object image are recovered by

$$\hat{\mathcal{F}}_1 = \mathcal{B}^{-1}(\hat{\mathbf{b}}_1) = (\hat{\xi}_1, \hat{\eta}_1, \hat{\xi}_2, \hat{\eta}_2), \quad (7)$$

and

$$\hat{\mathcal{F}}_2 = \Upsilon(\mathcal{B}^{-1}(\hat{\mathbf{b}}_2)), \quad (8)$$

where  $\mathcal{B}^{-1}(\cdot)$  is the inverse binary mapping and  $\Upsilon(\cdot)$  denotes the upsampling operation. The background image index is recovered as

$$\hat{\alpha} = \mathcal{B}^{-1}(\hat{\mathbf{a}}), \quad (9)$$

and the corresponding background image is retrieved from the knowledge base:

$$\hat{\mathcal{U}} = \hat{\mathcal{U}}_{\hat{\alpha}}. \quad (10)$$

The reconstructed object region is denoted by

$$\Omega = \{(\xi, \eta) | \hat{\xi}_1 \leq \xi \leq \hat{\xi}_2, \hat{\eta}_1 \leq \eta \leq \hat{\eta}_2\}. \quad (11)$$

Thus, the final reconstructed image is

$$\hat{\mathcal{X}}(\xi, \eta, \zeta) = \begin{cases} \hat{\mathcal{F}}_2(\xi - \hat{\xi}_1, \eta - \hat{\eta}_1, \zeta), & (\xi, \eta) \in \Omega, \\ \hat{\mathcal{U}}(\xi, \eta, \zeta), & \text{otherwise,} \end{cases} \quad (12)$$

where  $\zeta$  denotes the color channel index. This reconstruction strategy explicitly preserves the object target while using knowledge-base priors to recover non-critical background regions.

#### E. Semantic Compression Ratio

The proposed SemCom framework improves effective data rate through avoiding direct transmission of the full-resolution image. Let  $B_{\text{img}}$  denote the number of bits required to transmit the original image  $\mathcal{X}$  and  $B_{\text{sem}}$  denote the number of bits in the semantic stream  $[\mathbf{b}_1, \mathbf{b}_2, \mathbf{a}]$ . As the size of  $[\mathbf{b}_1, \mathbf{a}]$  is negligible compared to that of  $\mathbf{b}_2$ , the semantic compression ratio is approximately

$$\mu = \frac{B_{\text{img}}}{B_{\text{sem}}} = \frac{HWC}{H_s W_s C_s}, \quad (13)$$

where  $(H, W, C)$  are the original image dimensions, and  $(H_s, W_s, C_s)$  are those of the cropped and downsampled object region. As a result, a higher  $\mu$  yields a higher effective transmission rate and lower latency, directly benefiting real-time applications in UAV-enabled networks.

### III. THREE-LAYER ACROSS-FUSION-AIDED YOLO

This section presents the proposed three-layer AF-aided YOLO model, which improves the localization accuracy of object targets. As illustrated in Fig. 2, the detector consists of a backbone, a neck, and a detection head. Compared with the YOLO11 architecture, our model modifies the neck by incorporating the AF module, the duo-attention (DuoAtt) module, and the Hadamard-based feature enhancement (HFE) module.

#### A. Across-Fusion Module

The AF module is designed to enhance multi-scale feature interaction. Let  $\mathcal{Q}_1 \in \mathbb{R}^{C_1 \times H_1 \times W_1}$ ,  $\mathcal{Q}_2 \in \mathbb{R}^{C_2 \times H_2 \times W_2}$ ,  $\mathcal{Q}_3 \in \mathbb{R}^{C_3 \times H_3 \times W_3}$  denote the large-scale, middle-scale, and small-scale feature maps, respectively. The fused feature is obtained by

$$\mathcal{A} = \text{Concat}(P_{\text{max}}(\mathcal{Q}_1) + P_{\text{avg}}(\mathcal{Q}_1), \mathcal{Q}_2, \Upsilon(\mathcal{Q}_3)), \quad (14)$$

where  $P_{\text{max}}(\cdot)$  and  $P_{\text{avg}}(\cdot)$  denote adaptive max-pooling and average-pooling operations, respectively, and  $\text{Concat}(\cdot)$  represents channel-wise concatenation.

By jointly exploiting high-level semantic information, middle-level contextual information, and low-level spatial details, the AF module strengthens the representation of small

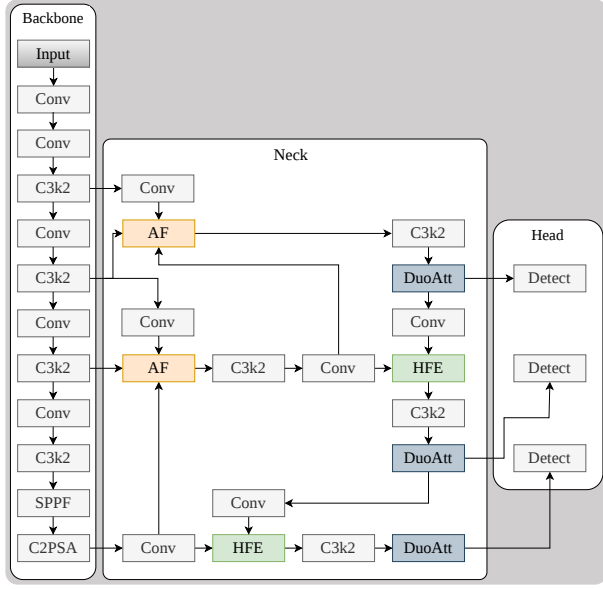


Fig. 2 Architecture of the proposed three-layer across-fusion (AF) aided YOLO detector.

and distant object targets. This is particularly beneficial for UAV-based outdoor localization, where targets often occupy only a small portion of the captured image.

#### B. Duo Attention Module

To enhance feature discrimination, the DuoAtt module is integrated into the neck [6]. This module captures cross-dimensional interactions along the channel–height and channel–width axes via two parallel attention branches, each computing attention weights through a sequence of rotation, pooling, convolution, and sigmoid activation operations. The resulting attention maps are aggregated to produce refined feature representations.

In contrast to conventional channel-only or spatial-only attention mechanisms, DuoAtt captures richer inter-dimensional dependencies with marginal computational overhead, thereby improving localization accuracy while preserving practical inference efficiency.

#### C. Hadamard-Based Feature Enhancement

The HFE module is introduced to strengthen the interaction between feature representations extracted from two different branches via element-wise multiplicative fusion. Given two input feature maps  $\mathcal{I}_1 \in \mathbb{R}^{C_0 \times H_0 \times W_0}$  and  $\mathcal{I}_2 \in \mathbb{R}^{C_0 \times H_0 \times W_0}$  with identical spatial and channel dimensions, the HFE output is calculated as

$$\mathcal{H} = \text{Conv}_{3 \times 3}(\mathcal{I}_1) \odot \text{Conv}_{3 \times 3}(\mathcal{I}_2), \quad (15)$$

where  $\text{Conv}_{3 \times 3}(\cdot)$  denotes a  $3 \times 3$  convolution operation, and  $\odot$  denotes the element-wise multiplication.

Unlike conventional concatenation-based fusion, our Hadamard-based strategy explicitly emphasizes correlated and mutually informative responses between the two

TABLE I: Detection Performance Comparison

| Model         | mAP@50       | mAP@50:95    | Inference Time (ms) |
|---------------|--------------|--------------|---------------------|
| YOLO11        | 0.918        | 0.553        | 8.9                 |
| TriAtt-YOLO   | 0.908        | 0.554        | 11.0                |
| AF-Aided YOLO | <b>0.923</b> | <b>0.581</b> | 12.5                |

feature branches, while suppressing weakly related or less discriminative components. As a result, the HFE module enhances shared salient representations and improves the effectiveness of feature fusion.

#### IV. PERFORMANCE ANALYSIS

Experiments are conducted on the SARD dataset [7], which comprises 5755 outdoor images for object target localization, each with a resolution of  $640 \times 640$  pixels. The dataset is partitioned into training, validation, and testing sets consisting of 4041, 1144, and 570 images, respectively.

##### A. Effective Detection

The detector is trained on an NVIDIA RTX 3050 GPU. The model is optimized using AdamW [8] with a learning rate of  $2 \times 10^{-3}$  and a weight decay of  $5 \times 10^{-4}$ . Training is performed for 100 epochs with a batch size of 8.

Table I compares the detection performance of various YOLO-based models, where  $\text{mAP}@ \phi$  denotes the mean average precision calculated at an intersection over union threshold of  $\phi\%$ . The proposed three-layer AF-aided YOLO achieves the highest  $\text{mAP}@50$  and  $\text{mAP}@50:95$ . Specifically, compared with YOLO11, the proposed model improves  $\text{mAP}@50$  from 0.918 to 0.923 and  $\text{mAP}@50:95$  from 0.553 to 0.581, while maintaining a practical inference time of 12.5 ms.

The improvement demonstrates that the proposed AF module, together with the DuoAtt and HFE modules, effectively enhances multi-scale feature fusion. More accurate target localization provides reliable spatial constraints for semantic encoding and contributes to higher reconstruction fidelity at the receiver.

##### B. Reliable Reconstruction

To evaluate the image reconstruction performance over Rician fading channels, Fig. 3 compares the proposed method with representative baselines, including DeepJSCC [9], DynamicJSCC [10], and JPEG+LDPC [11]. The performance is assessed in terms of peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and learned perceptual image patch similarity (LPIPS). The Rician factor is set to  $\kappa = 10$  dB, which is a typical value for UAV-enabled wireless networks [12]. At the physical layer, low-density parity-check (LDPC) coding with code rate  $2/3$  and BPSK modulation are adopted. Moreover, MRC with  $M$  diversity branches is applied to the transmission of critical semantic streams  $\mathbf{b}_1$  and  $\mathbf{a}$ .

As shown in Fig. 3, the proposed SemCom framework consistently achieves higher PSNR and SSIM, as well as lower LPIPS, than the benchmark schemes, over the entire SNR range. This improvement is mainly attributed to two aspects:

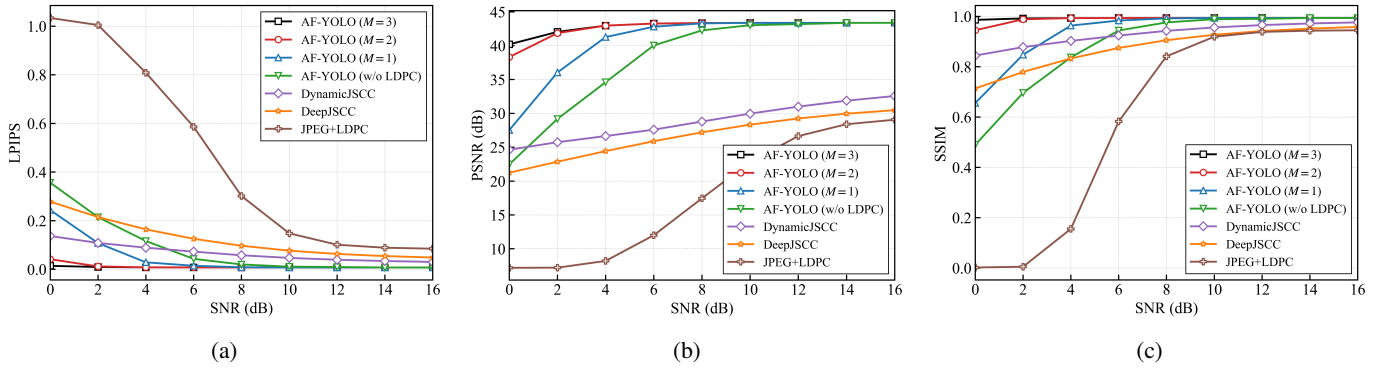


Fig. 3 Image reconstruction performance versus SNR over Rician channels: (a) LPIPS, (b) PSNR, (c) SSIM.

TABLE II: Transmitted Content and Compression Ratio

| Method      | Transmitted Content      | Compression Ratio |
|-------------|--------------------------|-------------------|
| JPEG+LDPC   | Compressed image + LDPC  | 2.4               |
| DeepJSCC    | Latent image features    | 6                 |
| DynamicJSCC | Adaptive latent features | [6, 12]           |
| Proposed    | Target semantics + index | 167               |

i) The proposed AF-aided YOLO detector enables accurate target localization, thereby preserving the structural fidelity of the reconstructed region of interest. ii) The shared background image knowledge base provides reliable background priors at the receiver, effectively mitigating background distortion caused by wireless channel impairments. The performance gain is particularly pronounced in the low-SNR regime, where conventional transmission schemes suffer from severe reconstruction degradation.

Furthermore, the diversity gain provided by MRC substantially enhances the reliability of critical semantic information transmission. In particular, a small diversity order, e.g.,  $M = 2$ , is sufficient to achieve a favorable tradeoff between robustness improvement and implementation complexity.

### C. Efficient Transmission

Table II summarizes the transmitted content and compression ratio comparisons. Since the proposed method transmits only the bounding-box coordinates, cropped object image, and background index, the number of transmitted bits is substantially reduced compared with traditional image transmission, e.g., JPEG+LDPC. This confirms that the proposed semantic compression strategy improves transmission efficiency by eliminating redundant background transmission. The background information is not repeatedly transmitted but retrieved from the shared knowledge base, which is particularly suitable for UAV-enabled networks with limited spectrum and energy resources.

## V. CONCLUSION

This letter proposed a mission-oriented SemCom framework for UAV-enabled wireless networks. By transmitting only task-critical semantic information and recovering background regions from a shared knowledge base, the proposed

method improves the semantic compression ratio and effective transmission rate. A three-layer AF-aided YOLO detector was developed to enhance target object localization accuracy, thereby improving the structural consistency and quality of reconstructed images. Moreover, MRC was employed for the critical coordinate stream to enhance transmission reliability over Rician fading channels. Simulation results demonstrated that the proposed framework achieves superior reconstruction quality, higher transmission efficiency, and improved wireless robustness compared with representative baselines.

## REFERENCES

- [1] M. Hui, *et al.*, "UAV-assisted mobile edge computing: Optimal design of UAV altitude and task offloading," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 13633-13647, Oct. 2024.
- [2] Y. Sun, *et al.*, "Multi-scale enhanced semantic communication for robust multi-resolution image transmission," *IEEE Wireless Commun. Lett.*, vol. 14, no. 11, pp. 3819-3823, Nov. 2025.
- [3] L. Qiao, *et al.*, "Latency-aware generative semantic communications with pre-trained diffusion models," *IEEE Wireless Commun. Lett.*, vol. 13, no. 10, pp. 2652-2656, Oct. 2024.
- [4] H. Hu, *et al.*, "Resource allocation for multi-modal semantic communication in UAV collaborative networks," *IEEE Trans. Commun.*, vol. 73, no. 9, pp. 7599-7616, Sep. 2025.
- [5] L. Bai, *et al.*, "Computer vision-based localization with visible light communications," *IEEE Trans. Wireless Commun.*, vol. 21, no. 3, pp. 2051-2065, Mar 2022.
- [6] D. Misra, T. Nalamada, A. U. Arasanipalai, and Q. Hou, "Rotate to attend: Convolutional triplet attention module," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3139-3148.
- [7] N. Gegenava, "SARD: Search and Rescue Dataset," Kaggle. [Online]. Available: <https://www.kaggle.com/datasets/nikolasgegenava/sard-search-and-rescue>. [Accessed: Jun. 21, 2026]
- [8] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," in *Proc. ICLR*, 2019, pp. 1-8.
- [9] E. Bourtsoulatz, D. B. Kurka and D. Gündüz, "Deep joint source channel coding for wireless image transmission," *IEEE Trans. Cognit. Commun. Netw.*, vol. 5, no. 3, pp. 567-579, Sep. 2019.
- [10] M. Yang and H. -S. Kim, "Deep joint source-channel coding for wireless image transmission with adaptive rate control," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP)*, May 2022, pp. 5193-5197.
- [11] N. Thomos, N. V. Boulgouris and M. G. Stintzis, "Optimized transmission of JPEG2000 streams over wireless channels," *IEEE Trans. Image Process.*, vol. 15, no. 1, pp. 54-67, Jan. 2006.
- [12] J. Rodríguez-Piñeiro, *et al.*, "Air-to-Ground channel characterization for low-height UAVs in realistic network deployments," *IEEE Trans. Antennas Propag.*, vol. 69, no. 2, pp. 992-1006, Feb. 2021.